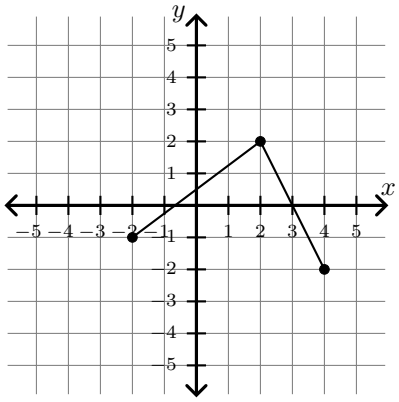
Section 6.6 — Function Transformations

Math 103: College Algebra

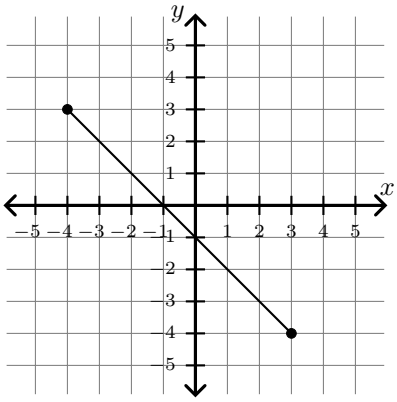
1 Single-Step Transformations

In each figure below, the given graph is the graph of $y = f(x)$. Graph the function in the question on the same axes. You may find it helpful to use the marked points as reference points.

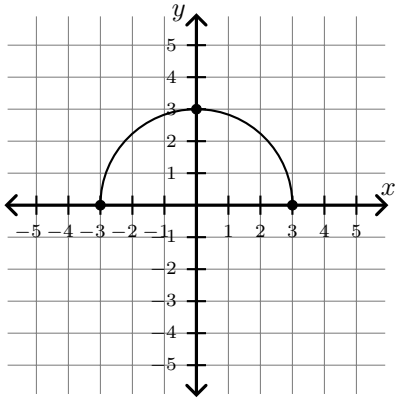
1. $g(x) = 2f(x)$



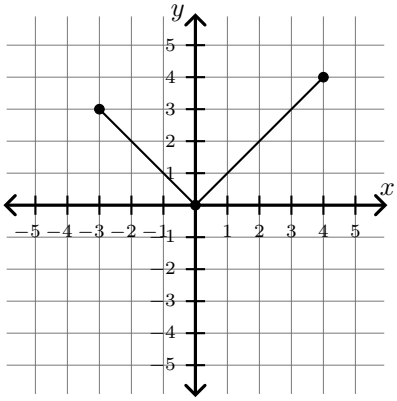
2. $g(x) = f(-x)$



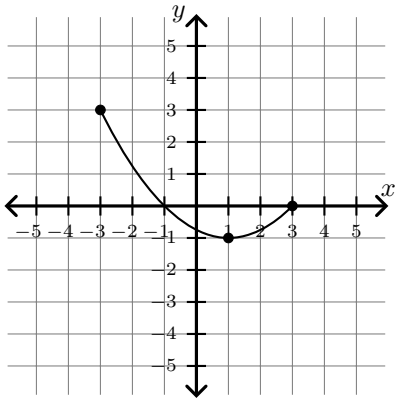
3. $g(x) = f(x) - 3$



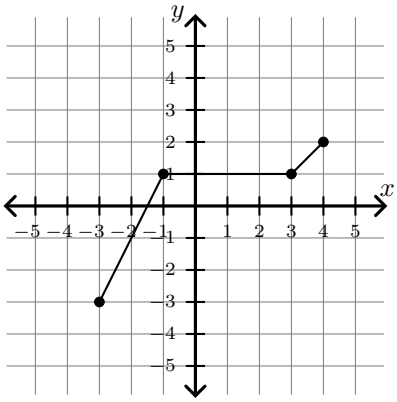
4. $g(x) = -f(x)$



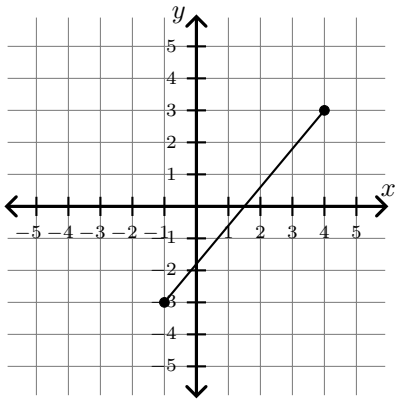
5. $g(x) = f(x) + 2$



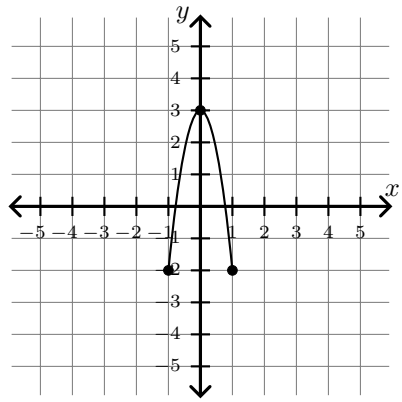
6. $g(x) = f(x + 1)$



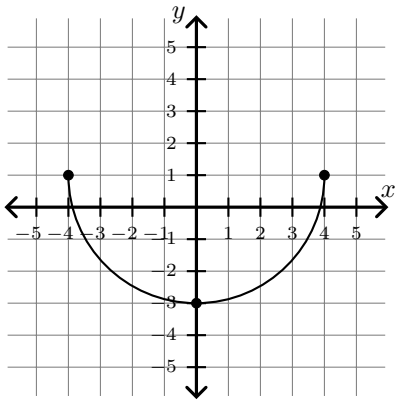
7. $g(x) = \frac{2}{3}f(x)$



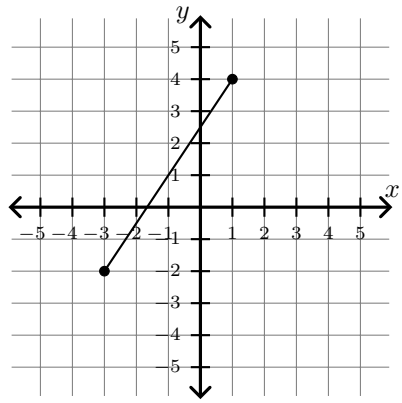
8. $g(x) = f(\frac{1}{3}x)$



9. $g(x) = f(2x)$



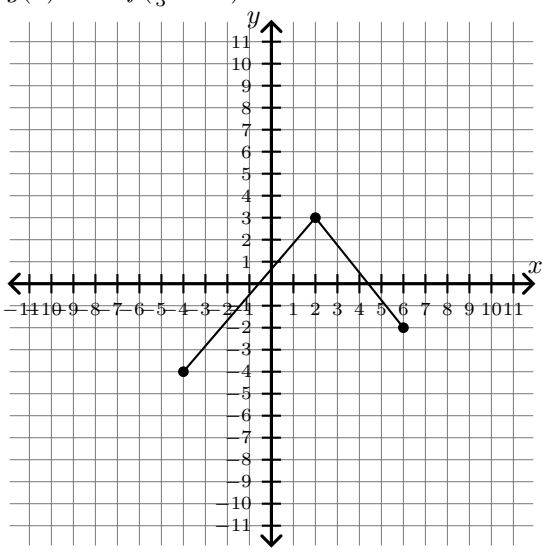
10. $g(x) = f(x-3)$



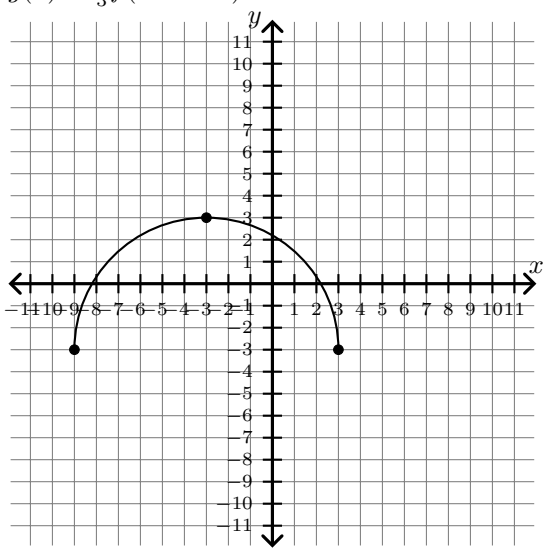
2 Multiple Transformations

In each figure below, the given graph is the graph of $y = f(x)$. Graph the function in the question on the same axes. You may find it helpful to use the marked points as reference points. *You will want to write out your work on a separate sheet of paper.*

11. $g(x) = -2f(\frac{2}{3}x + 2) + 3$



12. $g(x) = \frac{2}{3}f(-2x - 5) - 1$



3 Domain and Range

13. If $g(x) = 3f(-2x-3)+4$ and $\operatorname{Dom} f = (-5,\infty)$, what is $\operatorname{Dom} g$?

14. If $h(x) = \frac{1}{3}g(x+1)-7$ and $\operatorname{Rng} g = (-\infty,3]\cup[5,\infty)$, what is $\operatorname{Rng} h$?

15. If $k(x) = 2f\left(7-\frac{2}{5}x\right)-6$, $\operatorname{Dom} f = (0,1)$, and $\operatorname{Rng} f = (-\infty,\infty)$, what are $\operatorname{Dom} k$ and $\operatorname{Rng} k$?

4 Symmetry

Classify each function as even, odd, or neither.

16. $f(x) = x^2$
17. $f(x) = (x-7)^2$
18. $f(x) = x^5$
19. $f(x) = 2x$
20. $f(x) = x^6 - 3x^2 + 9$
21. $f(x) = x^3 + x^2$
22. $f(x) = x^3 + 6$
23. $f(x) = x^9 - x^3 + 8x$
24. $f(x) = \sqrt{x}$
25. $f(x) = \sqrt[5]{x}$
26. $f(x) = 3|2x| - 5$
27. $f(x) = |x-5|$
28. Based on your answers to #16 – 23, how can we decide if a polynomial is an even function, an odd function, or neither? (Try to answer this question on your own before looking at the answer key.)

5 Using the Library

For each function below:

a. Identify the base function (call it $f(x)$) and the family of functions it belongs to.

b. Rewrite the function in terms of $f(x)$. (We usually won't do this explicitly in practice.)

c. Sketch a graph of the function, including any asymptotes and clearly marking any special points, using a separate sheet of graph paper or the other side of this page.

d. Give the domain and range of the function.

29. $g(x) = -3x + 5$
30. $g(x) = \dfrac{5}{x^2 - 6x + 9}$
31. $g(x) = \dfrac{3}{x-2}$
32. $g(x) = |x-5| + 2$
33. $g(x) = 4$
34. $g(x) = 2(x-1)^3 + 1$
35. $g(x) = \log_{1/3}(2x+3) + 1$
36. $g(x) = -\left(\frac{1}{2}\right)^{x+2} + 1$
37. $g(x) = 2(x-2) + 1$
38. $g(x) = -3(x+4)^2 + 2$
39. $g(x) = \dfrac{3}{2x^2} - 1$
40. $g(x) = 3\sqrt[3]{4-x} - 1$
41. $g(x) = \dfrac{6x-7}{2x-3}$
42. $g(x) = 2\sqrt{-\frac{1}{2}x + 3}$
43. $g(x) = 2x^2 - 8x + 5$
44. $g(x) = 3^{\frac{1}{3}x} - 5$
45. $g(x) = \log_2(x-3) + 4$
46. $g(x) = -\left|4 - \frac{1}{3}x\right| + 2$